

Bazil Uddin khan

24k-0559

BSCS-1H

ICT LAB 11

Q1).

TASK1:

main.cpp	Output
<pre>1 #include <iostream> 2 #include<string.h> 3 using namespace std; 4 class Animal { 5 private : 6 string Name; 7 int Age; 8 public : 9 Animal(string name,int age) 10 { 11 Name = name; 12 Age = age; 13 } 14 15 void setName(string name) 16 { 17 Name = name; 18 } 19 void setAge(int age) 20 { 21 Age = age; 22 } 23 string GetName() 24 { 25 return Name; 26 }</pre>	<pre>Enter Name ::Giraffe Enter Age ::9 Name is : Giraffe Age is :9 === Code Execution Successful ===</pre>

TASK2:

main.cpp	Output
<pre>30 return Age; 31 } 32 void displayinfo(string name,int age) 33 { 34 35 cout<<"Name is :"<<name<<endl; 36 cout<<"Age is :"<<age<<endl; 37 } 38 39 40 41 }; 42 43 int main() 44 { 45 46 string name; 47 int age; 48 49 cout<<"Enter Name ::"; 50 cin>>name; 51 cout<<"Enter Age ::"; 52 cin>>age; 53 Animal info(name,age); ..</pre>	<pre>Enter Name ::Giraffe Enter Age ::9 Name is :Giraffe Age is :9 === Code Execution Successful ===</pre>

TASK3,TASK4:

main.cpp	Output
<pre>1 #include <iostream> 2 #include<string.h> 3 using namespace std; 4 class Animal { 5 private : 6 string Name; 7 int Age; 8 public : 9 Animal(string name,int age) 10 { 11 Name = name; 12 Age = age; 13 } 14 15 void setName(string name) 16 { 17 Name = name; 18 } 19 void setAge(int age) 20 { 21 Age = age; 22 } 23 string GetName() 24 { 25 return Name; 26 }</pre>	<pre>Enter Name ::Lion Enter Age ::9 Name is : Lion Age is : 9 Mammal information Name is : Lion Age is : 9 Feeding baby mammal Birds information Name is : Lion Age is : 9 Laying an egg. Reptile information Name is : Lion Age is : 9 Shedding skin.</pre>

Q1 CODE

```
#include <iostream>
```

```
#include<string.h>
```

```
using namespace std;
```

```
class Animal {
```

```
private :
```

```
    string Name;
```

```
    int Age;
```

```
public :
```

```
    Animal(string name,int age )
```

```
{
```

```
    Name = name;
```

```
    Age = age;
}

void setName(string name)
{
    Name = name;
}

void setAge(int age)
{
    Age = age;
}

string GetName()
{
    return Name;
}

int GetAge()
{
    return Age;
}

void displayinfo(string name,int age)
{

    cout<<"Name is : \n"<<name<<endl;
    cout<<"Age is : \n"<<age<<endl;
}
```

```
};  
  
class Mammal : public Animal{  
public :  
    void feedBaby()  
    {  
        cout<<"Feeding baby mammal \n";  
    }  
  
    Mammal(string name,int age ) :Animal(name,age)  
    {  
    }  
  
};
```

```
class Bird: public Animal{  
public :  
    void layEgg()  
    {  
        cout<<"Laying an egg.\n";  
    }  
  
    Bird(string name,int age ) :Animal(name,age)  
    {  
    }  
  
};
```

```
class Reptile: public Animal{  
public :
```

```
void shedSkin()
{
    cout<<"Shedding skin.\n";
}

Reptile(string name,int age ):Animal(name,age)
{
}

};
```

```
int main()
{

    string name;
    int age;

    cout<<"Enter Name ::";
    cin>>name;
    cout<<"Enter Age ::";
    cin>>age;
    Animal info(name,age);
    Mammal mammal(name,age);
    Bird bird(name,age);
    Reptile reptile(name,age);

    info.displayinfo(info.GetName(),info.GetAge());
    mammal.setName(name);
    mammal.setAge(age);
```

```
bird.setName(name);
```

```
bird.setAge(age);
```

```
reptile.setName(name);
```

```
reptile.setAge(age);
```

```
cout<<"Mammal information \n";
```

```
mammal.displayinfo(name,age);
```

```
mammal.feedBaby();
```

```
cout<<"Birds information \n";
```

```
bird.displayinfo(name,age);
```

```
bird.layEgg();
```

```
cout<<"Reptile information \n";
```

```
reptile.displayinfo(name,age);
```

```
reptile.shedSkin();
```

```
return 0;
```

```
}
```

Q2).

TASK1:

main.cpp	Run	Output
<pre>1 #include <iostream> 2 #include<string.h> 3 using namespace std; 4 5 class Person{ 6 private : 7 8 string Name; 9 int Age; 10 11 public: 12 Person(string name,int age) 13 { 14 Name = name; 15 Age = age; 16 } 17 void SetName(string name) 18 { 19 Name = name; 20 } 21 22 string GetName() 23 { 24 return Name; 25 } 26 void SetAge(int age)</pre>		<pre>enter your name :Ali enter your age :11 --- Code Execution Successful ---</pre>

TASK2).

main.cpp	Run	Output
<pre>17 void SetName(string name) 18 { 19 Name = name; 20 } 21 22 string GetName() 23 { 24 return Name; 25 } 26 void SetAge(int age) 27 { 28 Age = age; 29 } 30 31 int GetAge() 32 { 33 return Age; 34 } 35 void displayinfo(string name,int age) 36 { 37 cout<<"your name is :"<<name<<endl; 38 cout<<"your age is :"<<age<<endl; 39 } 40 }; 41 42 int main()</pre>		<pre>enter your name :Ali enter your age :9 your name is :Ali your age is :9 === Code Execution Successful ===</pre>

TASK3&&TASK4:

main.cpp	Run	Output
<pre>1 #include <iostream> 2 #include<string.h> 3 using namespace std; 4 5 class Person{ 6 private : 7 8 string Name; 9 int Age; 10 11 public: 12 Person(string name,int age) 13 { 14 Name = name; 15 Age = age; 16 } 17 void SetName(string name) 18 { 19 Name = name; 20 } 21 22 string GetName() 23 { 24 return Name; 25 } 26 void SetAge(int age)</pre>	Run	<pre>enter your name :john enter your age :25 your name is :john your age is :25 Enter your employee id :1234 Enter your student id :5678 employee info your name is :john your age is :25 your employee id is :1234 student info your name is :john your age is :25 your studen id is :5678 === Code Execution Successful ===</pre>

Q2 CODE:

```
#include <iostream>
```

```
#include<string.h>
```

```
using namespace std;
```

```
class Person{
```

```
private :
```

```
    string Name;
```

```
    int Age;
```

```
public:
```

```
Person(string name,int age)
{
    Name = name;
    Age = age;
}
void SetName(string name)
{
    Name = name;
}

string GetName()
{
    return Name;
}
void SetAge(int age)
{
    Age = age;
}

int GetAge()
{
    return Age;
}
virtual void displayinfo(string name,int age)
{
    cout<<"your name is :"<<name<<endl;
    cout<<"your age is :"<<age<<endl;
}
};
```

```

class Employee :public Person{
private :
    int employeeID;

public:
    Employee(string name,int age,int employeeid) : Person(name,age),employeeID(employeeid)
    {
    }
    void Setemployeeid(int employeeid)
    {
        employeeID = employeeid;
    }

    int Getemployeeid()
    {
        return employeeID;
    }

    void displayinfo (string name,int age,int employeeid)
    {
        cout<<"your name is :"<<name<<endl;
        cout<<"your age is :"<<age<<endl;
        cout<<"your employee id is :"<<employeeID<<endl;
    }
};

```

```

class Student :public Person{
private :

```

```
int studentID;
```

```
public:
```

```
Student(string name,int age,int studentid) : Person(name,age),studentID(studentid)
```

```
{
```

```
}
```

```
void Setstudentid(int studentid)
```

```
{
```

```
    studentID = studentid;
```

```
}
```

```
int Getstudentid()
```

```
{
```

```
    return studentID;
```

```
}
```

```
void displayinfo(string name,int age,int studentid)
```

```
{
```

```
    cout<<"your name is :"<<name<<endl;
```

```
    cout<<"your age is :"<<age<<endl;
```

```
    cout<<"your studen id is :"<<studentID<<endl;
```

```
}
```

```
};
```

```
int main()
```

```
{
```

```
    string name;
```

```
    int age;
```

```
    int employeeid;
```

```
    int studentid;
```

```
cout<<"enter your name :";
```

```
cin>>name;
```

```
cout<<"enter your age :";
```

```
cin>>age;
```

```
Person person(name,age);
```

```
person.displayinfo(person.GetName(),person.GetAge());
```

```
cout<<"Enter your employee id :";
```

```
cin>>employeeid;
```

```
cout<<"Enter your student id :";
```

```
cin>>studentid;
```

```
Employee employee(name,age,employeeid);
```

```
Student student(name,age,studentid);
```

```
cout<<"employee info \n";
```

```
employee.displayinfo(name,age,employeeid);
```

```
cout<<"student info \n";
```

```
student.displayinfo(name,age,studentid);
```

```
return 0;
```

}